

Education

University of Colorado at Boulder	May 2026
B.S., Aerospace Engineering	Boulder, CO
Relevant Coursework: Medicine in Space Environments; Orbital Mechanics; Aircraft Dynamics; Dynamics Controls Lab; Electronics; Thermodynamics and Heat Transfer; Structures; Aerodynamics; Foundations of Propulsion	
Computer Science Minor	
- C/C++, MATLAB, and Python programming with low-level Assembly processes - Coursework: Algorithms; Data Structures; Computer Systems; Database Systems; Computational Methods	

Skills

Programming: C, C++, Python, MATLAB, MySQL
Software: Adobe Photoshop/Lightroom, GitHub, SolidWorks, OpenVSP, EAGLE, LaTeX/Overleaf, MATLAB Simulink
Hardware: Rapid Prototyping, 3D Printing, PCB Manufacturing, Woodworking, Machine Shop, Soldering

Technical Experience

- Retractable Canard Passenger Aircraft:
 - Using ArduPilot and Inav firmware on fixed wing prototype for flight control, telemetry, and video.
 - Prototyping retractable canard on sub-scale technological demonstrator to analyze stability behavior in the transient state.
 - Performed control analysis in MATLAB to determine control surface sizing and modal behavior.
 - Present paper at AIAA student conference and present work to professors in design review presentations.
- EVA Medical Decision Aid (Mars Desert Research Station)
 - Validated aid in an analog environment at the MDRS to evaluate performance in simulated conditions.
 - Developed a risk decision matrix by organizing NASA medical standards into a weighted model to quantify likelihoods.
 - Engineered an interface to reduce cognitive load, enabling medical triage and critical decision-making during EVAs.
- Arthritis Assisting Glove:
 - Designed and manufactured custom PCB from EAGLE CAD
 - Integrated flex sensors with an Arduino in a wearable glove for assisted finger movement
- Boost-glide project:
 - Modeled an air propelled bottle rocket using predictive simulation in MATLAB.
 - Validated against experimental data for existing gliders and aircraft.
 - Iteratively fabricated design in machine shop and performed a boost-glide test.

Work Experience

Epic Productions	May 2022 – Present
Lead Wedding Photographer	Boulder, CO
- Manage wedding projects through planning and execution, ensuring client requirements and deadlines were met. - Use Adobe and Microsoft tools for time management and project planning.	
City of Austin	May 2021 – Aug 2023
Head Lifeguard	Austin, TX
- Maintained emergency procedures and risk assessment as well as managing scheduling. - Instructed swim lessons for children and coached for youth swim team.	
Austin Tri-Cyclist	May 2019 - August 2022
Bike Mechanic	Austin, TX
- Managed retail of triathlon gear including restocking and inventory logistics. - Maintained, built, and sold bikes for triathlon athletes and mountain bike team. - Performed maintenance on high, delicate carbon fiber frames to specification.	

Leadership and Activities

AIAA	Nov 2024 – Present
Officer	Boulder, CO
- Organize campus networking events between students and Aerospace industry - Represent CU at conventions and industry events	
Westlake Video Crew	Aug 2019 – Dec 2021
Crew Head	Austin, TX
- Stage Manager for crew of 40+ members over 6 sold-out technical productions - Performed safety briefings and the coordination of a full orchestra, a dozen sub crews, and first responders	

Achievements

- 1st Place Freshman Projects Showcase: Arthritis Assisting Glove
- 1st Place UIL Broadcasting 2018: Westlake Video Crew